Trig 2 Homework

Review

1. $\left(\frac{4}{25}\right)^{-\left(\frac{3}{2}\right)}$
   $= \left(\sqrt{\frac{25}{4}}\right)^3$
   $= \left(\frac{5}{2}\right)^3$
   $= \frac{125}{8}$
2. $2\log_2 x - \log_2(x+3) = 2$
   $\log_2 \frac{x^2}{x+3} = 2$
   $\therefore \frac{x^2}{x+3} = 4$
   $x^2 - 4x - 12 = 0$
   $\therefore x = 6 \text{ or } x = -2$
   but $x > 0 \therefore x = 6$
3. $\theta = 392°, 572°$
4. (b): $y = -2(x+1)(x-1)^2$
5. $f'(x) = 4x - 3$
   $4x - 3 = 0 \therefore x = \frac{3}{4}$

Consolidation

1. $A = -224°, B = 34°, C = 394°$
2. a. $x = 12°, 128°$
   b. $x = 96$
   c. $x = 21°, 79°, 201°, 259°$
3. a. $\cos^2\theta = \frac{3}{4}$
   $\therefore \cos\theta = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$
   $\therefore \theta = 30, 150, 210, 330$
   b. $2\sin\theta\cos\theta + \sin\theta = 0$
   $\sin\theta(2\cos\theta + 1) = 0$
   $\cos\theta = -\frac{1}{2} \text{ or } \sin\theta = 0$
   $\theta = 120°, 240°$
   or
   $\theta = 180°$
   c. $\sqrt{3}\sin 3\theta = \cos 3\theta$
   $\sqrt{3} \cdot \tan 3\theta = 1$
   $\tan 3\theta = \frac{1}{\sqrt{3}}$
   $3\theta = 30°, 210°, 390°, 570°, 750°, 930° \ (0° \leq 3\theta \leq 1080°)$
   $\theta = 10°, 70°, 130°, 190°, 250°, 310°$

Exam Questions

Question 1: Solve $\tan 2x = -\sqrt{3}$ for $0 \leq x \leq 360$

We know that $\tan 60 = \sqrt{3}$.
Then $\tan -60 = -\sqrt{3}$.
$\therefore x = -30$. However, $0 \leq x \leq 360$.
$\Rightarrow x = 60°, 150°, 240°, 330°$

Question 2: Let $f(x) = 4x^3 + 4x^2 + 7x - 5$.

Show that $(2x - 1)$ is a factor of $f(x)$, and thus solve the equation

$$4\sin^3\theta + 4\sin^2\theta + 7\sin\theta - 5 = 0$$

for $0° \leq \theta \leq 360°$

By inspection, $f(x) = (2x - 1)(2x^2 + 3x + 5)$

$(2x^2 + 3x + 5)$ does not have real roots: the discriminant $b^2 - 4ac < 0$. (-31)

Thus when $f(x) = 0, x = \frac{1}{2}$

Let $x = \sin x$.
Then $f(\theta) = 4\sin^3\theta + 4\sin^2\theta + 7\sin\theta - 5$.
And therefore $\sin\theta = \frac{1}{2} \Rightarrow \theta = 30° \text{ or } 150°$